When Concepts Leak: Concept Bottleneck Models for Interpretable ECG Diagnosis on PTB-XL

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Code: github.com/AntonyBoza/ecg-concept-bottleneck-leakage

Abstract

Concept bottleneck models (CBMs) route a prediction through concepts a person can read, so that an expert can see why a diagnosis was made and fix a wrong concept at test time. We put that promise to the test on 12-lead ECG diagnosis with PTB-XL (21,388 records, 5 diagnostic superclasses), using 19 form and rhythm statements as concepts and leaving out the four that duplicate diagnostic labels.

With a shared 1D ResNet encoder, the black box reaches a macro AUROC of 0.924 ± 0.002, in line with the published benchmark. Independent and sequential CBMs drop to 0.819 and 0.827, and most of the loss falls on conduction disturbances (−0.25), which we had predicted before training. A joint CBM gets close to the black box (0.915). Give it the annotated concepts instead of its own predictions, though, and it falls to 0.679.

Correcting concepts made every soft CBM worse, and we trace this to two causes that can be separated. Form annotations are incomplete: correcting form concepts on 707 test records without form annotation costs ten times more AUROC than on 707 records that have it. And soft concept probabilities carry diagnostic information of their own: binarizing them costs 0.17 AUROC. A hard CBM closes that channel and behaves as a CBM should, improving with each correction, but it tops out near 0.70. Adding a residual channel trained with dropout brings accuracy back to 0.925 and keeps corrections helpful (+0.005 AUROC, 95% CI +0.003 to +0.007). The catch is that the concepts then carry only 1–3% of the decision.

1. Introduction

Deep networks now read ECGs about as well as cardiologists, but a clinician still cannot see why a diagnosis was made, or correct the model when it gets one wrong. Concept bottleneck models (Koh et al., 2020) try to fix both. The network first predicts concepts a person can read, and a simple head turns those concepts into a diagnosis. A clinician can inspect the concepts, overwrite a wrong one (an *intervention*), and the prediction should improve.

That last step quietly assumes two things. The concept labels have to be complete and correct, and the concepts have to be the only path from the signal to the diagnosis. Clinical data rarely guarantees either. We wanted to see what happens to a CBM when both assumptions meet a large public ECG dataset whose concepts come from routine clinical reports.

Three questions guided the work:

1. How much diagnostic performance does a CBM give up against an equivalent black box, and on which diagnoses?
2. Do test-time interventions with annotated concepts improve predictions, and does the order of correction matter?
3. Do the learned concept → diagnosis weights make clinical sense?

This report contributes:

- A comparison of independent, sequential, joint and hard CBMs against a black box that shares the same encoder and training protocol, over 5 seeds, with paired bootstrap tests on the official PTB-XL folds.

- A prediction written down after the exploratory analysis and before any model was trained: the CBM would lose most on conduction disturbances. It did, by 0.25 AUROC.
- Evidence that interventions hurt every soft CBM, the independent one included, and an experiment that pulls apart the two causes: incomplete annotations and information leaking through soft concept probabilities.
- A hard CBM that brings intervenability back and shows where the concept vocabulary tops out (about 0.70 macro AUROC), and a hybrid CBM with a residual channel that matches the black box while corrections still help.

2. Related work

Concept bottleneck models. Koh et al. (2020) introduced independent, sequential and joint training of CBMs and showed that interventions raise accuracy on curated datasets with complete concept labels. Concept embedding models (Espinosa Zarlenga et al., 2022) give up some interpretability for accuracy by representing each concept with a vector instead of a single probability.

Concept leakage. Mahinpei et al. (2021) showed that soft concept probabilities can encode task information unrelated to what the concepts mean, so a bottleneck can be accurate without being faithful. Havasi et al. (2022) proposed hard concepts and side channels to keep this in check. We look at the same problem on clinical data, where the concept labels are also incomplete.

Intervenability. Shin et al. (2023) compared intervention policies and found uncertainty-guided ordering effective when concept labels can be trusted. Marcinkevičs et al. (2024) defined intervenability for black-box models and fine-tuned them to respond to concept edits. Stochastic CBMs (Vandenhirtz et al., 2024) model dependencies between concepts, so that correcting one updates related ones. All of this work assumes that an annotated concept is more reliable than a predicted one. On PTB-XL, that assumption often fails.

ECG benchmarks. PTB-XL (Wagner et al., 2020) contains 21,799 clinical 12-lead ECGs annotated with SCP-ECG statements of three types: diagnostic, form and rhythm. Strodthoff et al. (2021) set deep-learning benchmarks on it, and we borrow their random 2.5 s cropping.

3. Data and concepts

We use PTB-XL v1.0.3 at 100 Hz (10 s, 12 leads). The task is multi-label classification into 5 diagnostic superclasses, and 21,388 of the 21,799 records have at least one. Prevalence ranges from 44.5% for NORM to 12.4% for HYP (MI 25.6%, STTC 24.5%, CD 22.9%).

Concept set. The concepts are the SCP-ECG form and rhythm statements. Four statements are both form and diagnostic (DIG, LNGQT, NDT, NST_), and all four belong to STTC. Keeping them would put the label inside the bottleneck, so we drop them. Of the 27 left, we keep the 19 with at least 40 positives in the training folds (12 form, 7 rhythm). Only training data is used for this choice.

Incomplete annotation. Rhythm statements cover 96.5% of records. Form statements cover only 33.5% (7,302 records), and PTB-XL does not record whether form was ever assessed, so a missing form code can mean "absent" or just "not written down". We follow common practice and treat it as negative; Section 6.4 measures what that costs.

Concept–diagnosis associations. Before training anything, we checked that the concepts made clinical sense by computing P(superclass | concept) (Figure 1). Q waves and abnormal QRS point to MI (0.80 and 0.83), ST depression and low T amplitude to STTC (0.74 and 0.84), and LVH voltage criteria to HYP (0.76). Only prolonged PR (LPR) points to CD (0.55). Bundle branch blocks, which make up most of CD, exist in PTB-XL only as diagnostic labels. That led to a prediction we recorded before training: a CBM would lose most on CD, and next most on STTC.

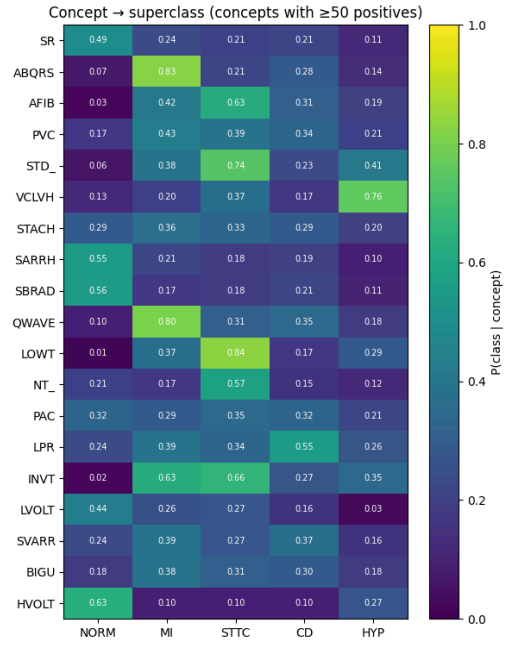


Figure 1. Concept → superclass association, P(superclass | concept present), for concepts with ≥ 50 positives. Rows do not sum to 1 (multi-label).

4. Methods

Encoder. Every model uses the same 1D ResNet: a stride-2 stem (kernel 15), four stages of two residual blocks (widths 64, 128, 192 and 256; kernel 9), and global average and max pooling concatenated into 512 features. It has 4.2 M parameters.

Black box. A linear layer with dropout maps the features to 5 logits, trained with binary cross-entropy.

CBMs. A concept head g maps the encoder features to K = 19 concept logits, and a linear task head f maps the concept probabilities $\hat{c} = \sigma(g(x))$ to the 5 superclass logits. We keep f linear on purpose, so its weights can be read as concept → diagnosis evidence. The four variants differ in how g and f are trained:

- **Independent:** g trained on concept labels; f trained on the *annotated* concepts c.
- **Sequential:** g as above; f trained on the *predicted* probabilities $\hat{c}$ of the training set.
- **Joint:** g and f trained end to end with the loss below ($\lambda = 1$).
- **Hard (sequential):** f trained on binarized predictions $1[\hat{c} > 0.5]$, so only the concepts' presence, not their soft values, reaches the head.

The joint loss is $L\text{joint} = \text{BCE}(f(\sigma(g(x))),\ y) + \lambda \cdot \text{BCE}(g(x),\ c)$, where y are the superclass labels and c the annotated concepts.

λ sweep. We also train the joint CBM with $\lambda \in \{0.1, 1, 10, 100\}$ (3 seeds each), to see how accuracy, concept fidelity and the response to interventions change with the concept-loss weight.

Hybrid (hard + residual). Here the task head reads the hard concepts plus a residual $r = W \cdot \varphi(x)$, a d-dimensional projection of the frozen black-box encoder features φ. Two controls stop the head from simply ignoring the concepts. Residual dropout zeroes r for a fraction p of training records, and a decorrelation penalty $\beta \cdot \text{mean}(\text{corr}(r, c)^2)$ discourages r from copying the concepts. Only W and the head are trained, on precomputed features averaged over windows.

Interventions. A simulated expert replaces the predicted values of k concepts with their annotated values, for k from 0 to 19. Concepts are corrected either in random order (averaged over 5 orders) or

by uncertainty, where each record's concepts with probability closest to 0.5 are fixed first.

5. Experimental setup

Splits and preprocessing. We use the official stratified folds: 1–8 for training (17,084 records), 9 for validation (2,146) and 10 for testing (2,158). Each lead is standardized with statistics from the training folds only. As in the benchmark, no filtering is applied.

Training. AdamW (learning rate 1e-3, weight decay 1e-2) with a one-cycle schedule, batch size 128 and mixed precision, for up to 60 epochs with early stopping (patience 10). Models see random 2.5 s crops during training. At test time we average outputs (for CBMs, the concept probabilities) over sliding 2.5 s windows with a 1.25 s stride. Checkpoints are chosen on validation macro AUROC: of the task for the black box and the joint CBM, of the concepts for the concept model. The test fold is never used for selection.

Decision rule for cropping. We adopted cropping only after it passed a threshold set before the run: validation AUROC above 0.925 on one seed, against 0.915 without it. It reached 0.928.

Statistics. Each model is trained with 5 seeds. We report the single-model mean ± standard deviation and the 5-seed ensemble with a 95% bootstrap confidence interval (1,000 resamples of test records). To compare two models we use a paired bootstrap on the same resampled records, and call a difference significant when its 95% interval excludes zero.

Compute. Everything ran on a single NVIDIA T4 GPU on Kaggle, at 4–5 minutes per seed. Code and notebooks: github.com/AntonyBoza/ecg-concept-bottleneck-leakage.

6. Results

6.1 Task performance

Every soft CBM loses to the black box, and how much depends on how the bottleneck is trained (Table 1). Independent and sequential CBMs give up about 0.10 macro AUROC. The joint CBM gives up only 0.009, a small gap but a significant one (paired bootstrap 95% CI −0.013 to −0.006).

Table 1. Test macro and per-class AUROC. Single model: mean ± std over 5 seeds; per-class values are single-model means. Ensemble: average of the 5 seeds, 95% bootstrap CI.

Model	Single model	Ensemble [95% CI]	NORM	MI	STTC	CD	HYP
Black box	**0.924 ± 0.002**	**0.933** [0.926–0.939]	0.947	0.921	0.936	0.917	0.901
CBM independent	0.819 ± 0.003	0.828 [0.818–0.838]	0.880	0.885	0.838	0.670	0.822
CBM sequential	0.827 ± 0.003	0.837 [0.827–0.847]	0.885	0.888	0.857	0.678	0.829
CBM joint	0.915 ± 0.003	0.923 [0.915–0.931]	0.945	0.912	0.931	0.911	0.877

The prediction we recorded in advance held up. Per class, the independent CBM loses 0.248 on CD, 0.098 on STTC, 0.079 on HYP, 0.067 on NORM and 0.035 on MI. CD falls to 0.670, not far above chance, while MI, which the Q-wave, QRS and T-wave concepts cover well, loses least. Random cropping on its own added 0.014 AUROC to the black box (0.910 → 0.924; ensemble gain +0.009, CI +0.006 to +0.012).

6.2 Concept prediction

The concept model predicts the concepts well, with a mean test AUROC of 0.922 for form and 0.944 for rhythm concepts. The joint CBM does worse on them (0.873 and 0.890), and sinus arrhythmia (SARRH) falls from 0.877 to 0.669. Part of the joint model's task accuracy is paid for by letting its concepts drift.

6.3 Interventions hurt every soft CBM

We expected interventions to help. They did not. Replacing predicted concepts with annotated ones lowers task AUROC for all three soft CBMs, and correcting the most uncertain concepts first makes things worse faster than correcting them at random (Figure 2). With all 19 concepts replaced, AUROC ends at 0.703 (independent), 0.699 (sequential) and 0.679 (joint). For the joint CBM, five corrections chosen by uncertainty already cost 0.20 (0.915 → 0.716).

The independent CBM is the most revealing case. Its head was trained on annotated concepts, yet it does better with predicted concepts (0.819) than with the annotations it was trained on (0.703). The predictions, in other words, carry more diagnostic information than the labels. The joint CBM goes further: with its own predicted concepts it reaches 0.911 on CD, against 0.548 with the annotated ones. The concept set cannot explain that CD performance, so the information has to travel through the soft probabilities.

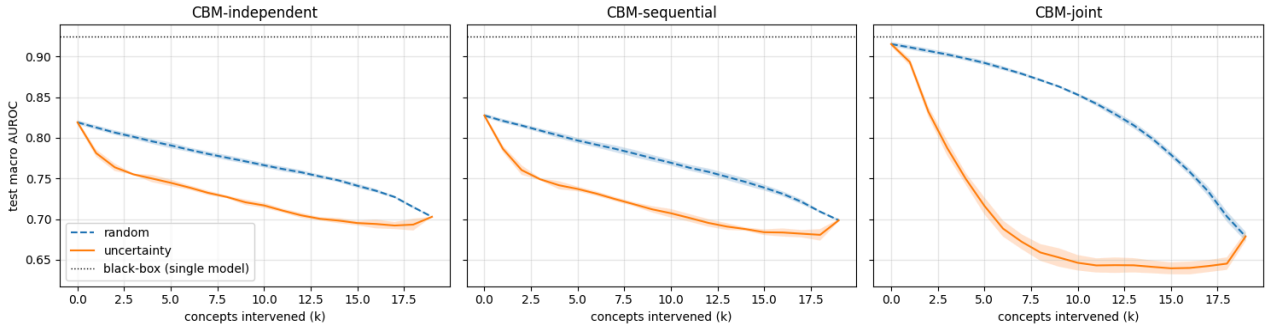


Figure 2. Test-time interventions. Mean ± std over 5 seeds; dotted line: black box (single model).

6.4 Why interventions hurt

Two mechanisms could explain this: incomplete form annotations (H1) and task information carried by the soft probabilities (H2). Table 2 changes one thing at a time to tell them apart.

Table 2. Test macro AUROC under controlled substitutions (mean of 5 seeds). Rows b–f change one thing relative to row a.

Condition	Independent	Sequential	Joint
a) Predicted concepts (soft)	0.819	0.827	0.915
b) Predicted concepts binarized at 0.5	0.647	0.647	0.805
c) Annotated rhythm concepts only	0.808	0.814	0.886
d) Annotated form concepts, all records	0.753	0.745	0.848
e) Annotated form, 707 form-annotated records	0.815	0.817	0.891
e′) Annotated form, 707 random non-annotated records	0.780	0.782	0.897
f) All concepts annotated	0.703	0.699	0.679

H2 is the larger effect. Binarizing the predicted concepts (row b) costs 0.17 AUROC in the independent and sequential CBMs, more than switching to annotated concepts altogether (row f). A concept model trained only on concept labels still hides diagnostic information in how confident it is. Correcting only the rhythm concepts, whose annotations are almost complete, costs just 0.01 (row c).

H1 holds too. Correcting form concepts on 707 records without form annotation (row e′) costs the independent CBM 0.039, ten times what the same correction costs on 707 records that have it (0.004, row e). For the sequential CBM the figures are 0.045 and 0.010. Where an annotation exists, it mostly agrees with the prediction. Where it is missing, the "correction" erases a finding the model picked up and the annotator never wrote down. The joint CBM shows no such gap (−0.024 on annotated,

−0.019 on non-annotated records): its concepts have drifted far enough from their meaning that any substitution hurts.

6.5 A hard CBM restores intervenability

A sequential CBM whose head reads only binarized concepts finally behaves the way a CBM is supposed to (Figure 3). It starts lower (0.649), but every correction helps, and corrections guided by uncertainty help more than random ones (0.689 vs. 0.664 at k = 5). With all concepts annotated it reaches 0.696, the same ceiling the annotated concepts hit in Table 2 (row f). Removing the soft channel costs 0.18 AUROC and buys interventions that work.

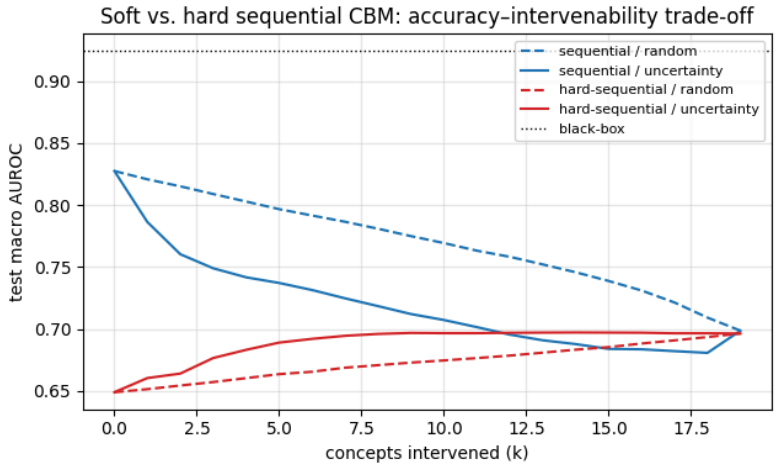


Figure 3. Accuracy–intervenability trade-off: soft vs. hard sequential CBM (mean of 5 seeds).

6.6 Concept → diagnosis weights

The independent CBM's linear head reads much like a cardiology textbook (Figure 4). MI is driven by abnormal QRS (+3.7), Q waves (+2.8) and T inversion (+1.5). HYP rests on LVH voltage criteria (+3.2) and ST depression (+1.4, the strain pattern), and STTC on ST depression, low T amplitude and T-wave changes (+1.5 to +1.8). High voltage on its own adds only +0.7 toward HYP, and low voltage counts against it (−0.5). CD has a single positive driver, prolonged PR (+1.4), and that gap in the vocabulary is what costs it 0.25 AUROC. These weights can only be read this way because the head was trained on annotated concepts. In the leaking models, the same numbers would not mean what their labels say.

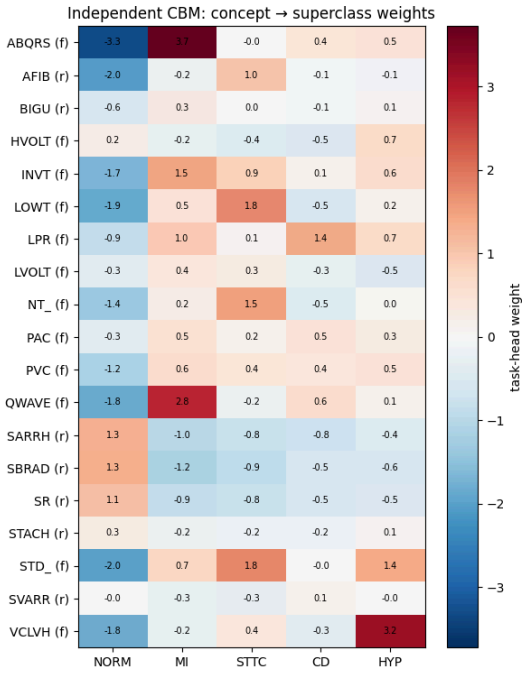


Figure 4. Concept → superclass weights of the independent CBM (mean of 5 seeds); (f) form, (r) rhythm concept.

6.7 No concept-loss weight escapes the trade-off

Sweeping the joint CBM's concept-loss weight λ from 0.1 to 100 moves the model along the trade-off but never off it (Table 3, Figure 5). Raising λ brings concept fidelity back and costs task accuracy. At no value of λ does correcting concepts help.

Table 3. Joint CBM vs. λ (test, mean of 3 seeds). Δk5: change in task AUROC after correcting the 5 most uncertain concepts.

λ	Task AUROC	Concept AUROC	Δk5	All concepts annotated	CD AUROC
0.1	0.921	0.623	−0.081	0.550	0.920
1	0.916	0.878	−0.205	0.675	0.910
10	0.851	0.927	−0.115	0.701	0.730
100	0.821	0.925	−0.088	0.699	0.694

At λ = 0.1 the "concepts" barely predict their own labels (0.62) while the task is still solved: leakage in its purest form. The λ = 10 row is subtler. Its concepts are as faithful as those of the concept-only model (0.927), yet it still beats the sequential CBM overall (0.851 vs. 0.827) and on CD (0.730 vs. 0.678). So concept AUROC does not detect leakage. It measures how concept probabilities are ranked, and the leaked information lives in their magnitude.

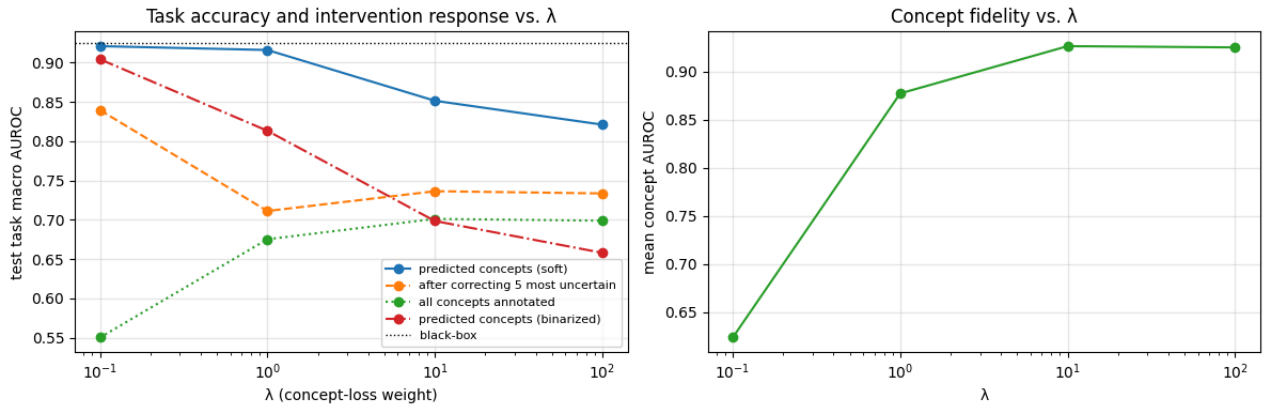


Figure 5. Joint CBM across λ: task accuracy and intervention response (left), concept fidelity (right).

6.8 A residual channel with dropout makes corrections safe

To get accuracy and safe interventions in the same model, we add a residual channel to the hard CBM: a d-dimensional projection of the black-box encoder features, concatenated with the hard concepts before the linear head. Two controls limit how far the head can bypass the concepts. Residual dropout zeroes the whole residual for a fraction p of training records, and a decorrelation penalty (weight β) discourages the residual from copying the concepts. We searched d ∈ {2, 8, 32}, p ∈ {0, 0.5} and β ∈ {0, 1} with 3 seeds. The configuration was picked by a rule fixed before the search and applied to validation data only: the most accurate one whose validation AUROC does not drop when concepts are corrected.

Table 4. Hybrid CBM (test, mean of 3 seeds). Δk5 / Δall: change after correcting the 5 most uncertain / all concepts.

Model	Task AUROC	Δk5	Δall	CD AUROC
Hard CBM (no residual)	0.645	+0.040	+0.047	0.573
Hybrid d = 2, p = 0.5, β = 1	0.887	+0.016	+0.017	0.916
Hybrid d = 32, p = 0, β = 0	0.925	+0.001	+0.001	0.920
Hybrid d = 32, p = 0.5, β = 1 (selected)	**0.925**	**+0.005**	**+0.006**	**0.921**

The selected hybrid matches the black box (0.925 vs. 0.924), and unlike any soft CBM it gets better when concepts are corrected: +0.0051 macro AUROC for the 5-seed ensemble (paired bootstrap 95% CI +0.0034 to +0.0067), with almost the same gain on every seed (±0.0001). Residual dropout is what makes the difference. Without it, corrections do essentially nothing (+0.001). The decorrelation penalty, on the other hand, moved results by less than 0.001 in every configuration we tried.

Residual capacity traces a clear frontier (Figure 6). The hard CBM gains most from corrections but is the least accurate. With d = 2 the model sits in between (0.887, +0.016), and with d = 32 it reaches black-box accuracy for a small gain.

The gain is real but small, and the reason is that the concepts carry very little of the decision. In the selected model the concept term explains only 1–3% of the variance of the class logits (NORM 2.2%, MI 1.4%, STTC 2.6%, HYP 1.1%) and 0.1% for CD. Described honestly, the hybrid is a black box whose concepts can be corrected without harm. It is not a model that explains its diagnoses through concepts.

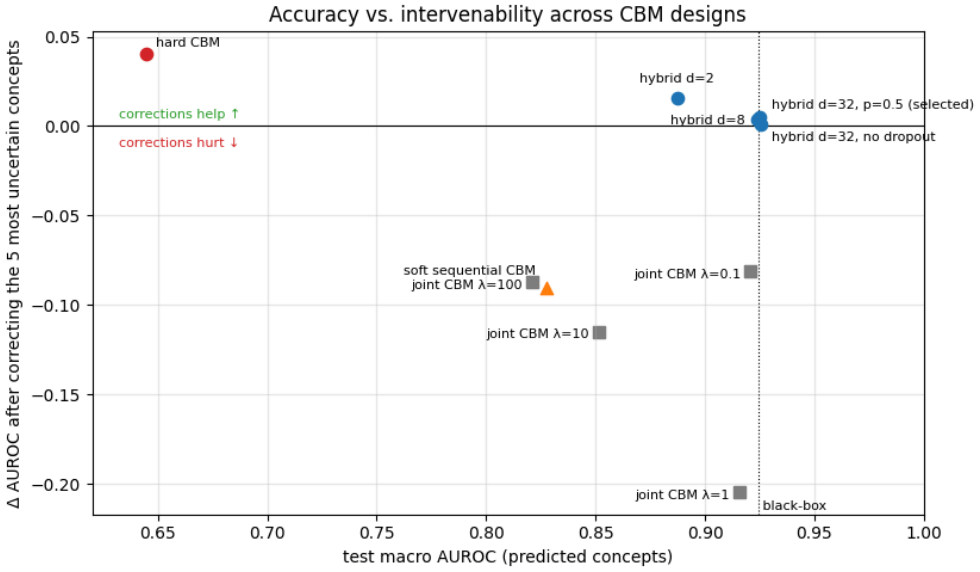


Figure 6. Accuracy vs. intervenability across CBM designs. Above the zero line, corrections help; dotted line: black box.

7. Discussion and limitations

Soft CBMs borrow part of their accuracy. On PTB-XL, what lets a soft CBM approach the black box is not what its concepts mean. It is the magnitude of the concept probabilities and, under joint training, concepts that slide away from their definitions. Someone reading such a model's concepts sees an explanation that looks faithful, for a prediction made on other grounds. To us this is the more worrying failure: not low accuracy, but a convincing explanation that is not true.

Interventions assume better labels than clinical data has. The CBM recipe treats annotated concepts as ground truth. In routine clinical reports, an absent finding is often just an unrecorded one, so the model can be right where the label is wrong. Benchmarks with complete, curated concept labels never expose this.

The concept vocabulary sets a ceiling. With annotated concepts and a linear head, macro AUROC stops at about 0.70, against 0.924 for the black box. Most of the gap sits in CD, whose defining findings (bundle branch blocks) exist in PTB-XL only as diagnostic labels. Richer concepts, such as QRS duration or bundle branch morphology, would raise the ceiling, provided they do not simply encode the diagnosis.

A residual channel buys accuracy and safety, not explanation. The hybrid shows that black-box accuracy and helpful corrections can coexist once the soft channel is gone and the residual is dropped out during training. But its concepts account for 1–3% of the logit variance, so it explains very little. On this data we could not get faithful concepts, high accuracy and concepts that actually drive the decision, all at once.

Limitations.

- Our "expert" is the dataset annotation. A cardiologist re-reading each ECG would probably supply more complete form labels and shrink the H1 effect.
- All results come from one dataset at one resolution (PTB-XL, 100 Hz). Datasets with complete concept labels may behave differently.
- The λ sweep and the hybrid grid use only 3 seeds and a few values each, and the hybrid reuses frozen black-box features instead of being trained end to end.
- The linear task head keeps the weights readable but cannot model interactions between concepts.

- We use PTB-XL v1.0.3, while the published benchmark used v1.0.1, so the comparison with it is only indicative.

8. Conclusion and future work

On clinical ECGs, a concept bottleneck model has to trade accuracy, faithfulness and intervenability against each other. Soft CBMs reach up to 0.915 macro AUROC, but they leak information and get worse when an expert corrects them. A hard CBM is faithful and correctable, and stops at 0.70. The 0.22 gap to the black box is information the clinical concept vocabulary does not hold. A residual channel with dropout closes that gap and keeps corrections helpful, but it leaves the concepts only 1–3% of the decision.

The hybrid shows that safe corrections and black-box accuracy can live in the same model. What remains open is how to make the concepts carry a real share of the decision without losing either. Our results point to three directions:

1. Concept-first residuals. Train the residual only on what the concepts cannot explain (for example, on the errors of the concept-only model), and report the concept share next to accuracy and intervention gain.
2. Annotation-aware interventions. Model the probability that a concept was annotated at all, so corrections are trusted where annotation is complete and weighted down where it is missing.
3. Concept dependencies. Use stochastic or structured concept models, so that correcting one finding, ST depression for example, updates related findings and the diagnosis coherently.

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